

Proxmox Storage DRS -- Operator Manual

Installing, configuring and running it safely

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What you need

- **Proxmox VE 9.2**, or a management host with network access to a PVE 9.x cluster's API (tcp/8006) — the tool does not need to run on a cluster member.
- A **Prometheus** already receiving PVE's per-disk `blockstat` metrics via the InfluxDB output plugin and Telegraf. If your cluster's dashboards already show per-VM disk I/O, this is already true; `pve-storage-drs verify-metrics` (page 2) confirms it before you rely on it. No new exporter or collector is required — see `IMPLEMENTATION_PLAN.md` section 3.1 if you want the detail of where the data comes from.
- A **PVE user or API token** with read access to VM and storage inventory and permission to call `move_disk`. An API token is preferred for unattended operation (`execution.mode: auto`); a username/password pair works for interactive use. See "Setting up the PVE credential" below for the exact privileges — the obvious-looking "just grant Audit everywhere" is not enough, and the gap is silent rather than an error.

Setting up the PVE credential

Grant, on **every storage the tool will manage** (the shared, image-holding storages in your groups, not backup targets):

- **Datastore.Audit and Datastore.Allocate** — not Audit alone. `GET /nodes/{node}/storage/{storage}/content` which the tool uses to cross-check each disk's real allocated size, silently returns an empty list under Audit-only permission: a clean 200 response with no content and no error, indistinguishable from "this storage genuinely has nothing on it" unless you already know to be suspicious. Audit alone is enough for every *other* read call the tool makes; this one endpoint is the exception, verified against a live PVE 9.2.11 cluster (see `IMPLEMENTATION_PLAN.md` section 3.5's note). Getting this wrong does not make the tool fail loudly — it makes it under-count what already occupies each storage, which quietly erodes the section 5.3 snapshot reserve it exists to protect.
- Grant it **at each storage's own path** (`/storage/<id>`), not only at the parent `/storage` — that is the grant confirmed to work.
- Also add `VM.Audit` cluster-wide (`/`) for VM inventory and config, which `PVEAuditor` already includes if you use that built-in role as a base.

If you are using an API token (recommended — see above), Proxmox's privilege separation means **the token has its own, separate ACL entries from its owning user**. A role granted to the user alone has no effect on the token, and a role granted to the token alone has no effect either unless the user has it too: the token's effective permission is the *intersection* of the two. Grant the same roles to both the user and `user@realm!tokenid`, or disable privilege separation on the token (Datacenter → Permissions → API Tokens → uncheck "Privilege Separation") so it always matches its user's permissions exactly.

A concrete, minimal setup: create a role (e.g. `pve-storage-drs`) with `Datastore.Allocate`, `Datastore.Audit`, `VM.Audit`, then add it as an ACL entry for both the user and the token at `/` for `VM.Audit`, and at each managed storage's `/storage/<id>` path for the datastore privileges.

Two further, VM-level privileges are needed once `apply` executes a `move` — `VM.Config.Disk` and `VM.Migrate`, both cluster-wide (`/`) or at least on every VM whose disks live in a managed group, granted to both the user and the token exactly as above. Neither is needed for anything this build actually runs today: `verify-metrics`, `show-load` and `verify-storages` only read, and only `move_disk` (called by `apply`, not yet implemented — `30-safety-and-status.md`) needs them. Grant them now alongside the privileges above so the credential does not need revisiting later; if you set up the credential read-only for now, add `VM.Config.Disk`, `VM.Migrate` to the role before you first run `apply`.

Installing the package

On a Debian trixie host (which is what Proxmox VE 9.x is built on):

```
apt install pve-storage-drs
```

This installs the `pve-storage-drs` executable, its manpage, the example configuration at `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml`, and this manual and the specification as PDFs under `/usr/share/doc/pve-storage-drs/`.

`coinor-cbc` and `python3-pulp` are Recommends, not Depends: without them the tool still plans, using the dependency-free heuristic of `IMPLEMENTATION_PLAN.md` section 5.5, but a Debian install should have them — `apt install pve-storage-drs` pulls them in by default.

Where the configuration lives

`/etc/pve/drs.yaml`. On a cluster member this path is on `pmxcfs`, the cluster filesystem, so the file is automatically the same on every node — there is exactly one copy and no question of which one is authoritative. A management host that is not itself a cluster member simply keeps an ordinary file at that same path.

Two things follow from that placement, and both matter before you put a password in the file:

- **An edit is live on every node the instant it is saved.** There is no staging step. This is why configuration mistakes are always hard failures (see [30-safety-and-status.md](#)) rather than warnings, and why `dry-run` is the default execution mode.
- **Files under `/etc/pve` are group-readable by `www-data`** — the PVE web server’s user. Check `ls -l /etc/pve/drs.yaml` on your cluster before deciding whether a plaintext password: is acceptable; the safer choice is an API token with `PVE_TOKEN_SECRET` set in the environment (or a `systemd` credential) rather than written into the file at all.

Copy `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml` to `/etc/pve/drs.yaml` and edit it — every key is commented with its default and what it does. The full reference is [10-configuration.md](#).

Where the configuration is *not*

`state.path` (default `/var/lib/pve-storage-drs/state.json`) is deliberately **not** on `/etc/pve`. It is rewritten on every run, `pmxcfs` writes need cluster quorum, and the state it holds — cooldown timestamps, the load vector at the last balance, the run lock — is meaningful only to the one host that is actually running the timer. See [10-configuration.md#statepath](#).

Running the timer on exactly one host

Putting the configuration on `pmxcfs` does not make the tool cluster-aware. `state.json` is node-local: each node would keep its own cooldowns, its own drift baseline and its own run lock, which do not coordinate with each other. **Enable the `systemd` timer on exactly one host.** The one thing that *does* cross the cluster is a startup scan for in-flight `move_disk` UPIDs owned by the DRS user, which is what keeps two accidental instances from actively conflicting (they degrade to “slow and redundant” instead).

First steps after installing

1. `pve-storage-drs -c /etc/pve/drs.yaml verify-metrics` — confirms the configured metric and label names actually exist in your Prometheus. See [20-verifying-metrics.md](#).
2. Read [10-configuration.md](#) and adjust the storage groups, thresholds and weights for your cluster.
3. See [30-safety-and-status.md](#) for exactly which commands this build supports end to end today, and which still report “not implemented yet”.

Configuration reference

Every key `pve-storage-drs` reads is documented here: type, unit, default, what it interacts with, and what happens if you set it too high or too low. A key that exists in the schema but not here, or here but not in the schema, is a bug — `tests/unit/test_documentation.py` asserts the two never drift apart.

`config/drs.example.yaml` in the source tree (installed to `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml`) carries the same defaults with inline comments; this page is the full explanation each comment points back to.

`schema_version`

Integer, required, no default.

The config format's version. `pve-storage-drs` refuses to start on a `schema_version` whose **major** number it does not understand, naming the version it does understand in the error, rather than guessing at fields from a future incompatible format. Currently 1.

proxmox — the cluster API connection

`proxmox.host`

String, required, no default.

Any cluster member's hostname or IP. The API is reached at `https://<host>:<port>/api2/json`.

`proxmox.port`

Integer, default 8006.

The PVE API port. Change this only if you have put something unusual (a reverse proxy) in front of the API.

`proxmox.verify_ssl`

Boolean, default `true`.

Whether to verify the API's TLS certificate. Set `false` only for a self-signed certificate you cannot otherwise trust, and prefer `proxmox.ca_file` instead where possible — disabling verification also disables protection against a compromised network path to every node's API.

`proxmox.ca_file`

String path or `null`, default `null`.

A CA bundle to verify the API's certificate against, when it is not signed by a certificate already trusted by the host running `pve-storage-drs`.

`proxmox.auth.username`

String or `null`.

A PVE user, e.g. `drs@pve`. Required unless `proxmox.auth.token_id` is set. Username/password authentication is what the original requirement asked for; an API token (below) is preferred for `execution.mode: auto` because it needs no interactive ticket refresh and can be scoped to exactly the privileges section 3.5 of the plan lists.

`proxmox.auth.password`

String or `null`, default `null`, environment: `PVE_PASSWORD`.

Leave this `null` in the file and set `PVE_PASSWORD` in the environment (or a systemd credential) instead, especially when the config lives on `/etc/pve` and is therefore group-readable by `www-data`. A password given in the file is used as-is and is **not** overridden by the environment variable — see [00-installation.md](#) for why the file's own permissions matter here.

proxmox.auth.token_id

String or null, default null.

An API token id, e.g. `drs@pve!balancer`. Set this instead of `username/password` for unattended operation; a token needs no ticket refresh loop and can be revoked independently of the user's password.

proxmox.auth.token_secret

String or null, default null, environment: `PVE_TOKEN_SECRET`.

The secret half of the API token, with the same environment-variable preference as `proxmox.auth.password`.

proxmox.ticket_refresh_seconds

Duration, default 3000 (50 minutes).

How often a username/password ticket is refreshed. PVE tickets last approximately two hours; refreshing well before that avoids a mid-run authentication failure. Not used for API-token authentication, which needs no ticket. Implemented by overriding proxmoxer's own (otherwise fixed-at-3600s) refresh interval after login, since proxmoxer has no constructor option for it; if a ticket somehow still gets rejected mid-run regardless (a long confirm-mode wait, a suspended process, a clock jump — see `50-pve-api.md`), the client transparently logs in again from scratch and retries the one call that failed, once, before giving up.

proxmox.read_workers

Integer, default 12.

Size of the bounded thread pool used to fetch per-VM configuration — `GET /nodes/{node}/qemu/{vmid}/config` has no batch form, so this is what keeps a several-hundred-VM cluster's topology read from being serial. Raising it trades API server load for wall-clock time; 8–16 is the range the plan suggests. Too high a value on a small PVE API server can itself become the bottleneck. Only the per-VM fetch is parallelized; the cluster-wide and per-storage calls (`50-pve-api.md`) still run once each, sequentially, before it starts.

prometheus — the metrics source**prometheus.url**

String, required, no default.

The base URL of your Prometheus, e.g. `http://prometheus.example.com:9090`.

prometheus.timeout_seconds

Duration, default 30.

Per-request timeout against Prometheus. Too low a value on a busy or long-range query makes `verify-metrics` and the load model fail spuriously; too high a value delays noticing that Prometheus is actually unreachable.

prometheus.username / prometheus.password

String or null, default null.

HTTP basic auth credentials, if your Prometheus requires them. Ignored when `prometheus.bearer_token` is set.

prometheus.bearer_token

String or null, default null.

A bearer token for Prometheus instances behind an auth proxy that expects one. Takes precedence over basic auth when both are set.

metrics — Telegraf/InfluxDB name mapping

Telegraf's naming is deployment-specific (IMPLEMENTATION_PLAN.md section 3.3), so every name below is configuration, never a constant. Run `pve-storage-drs verify-metrics` after changing any of them — see [20-verifying-metrics.md](#).

`metrics.read_ops`

String, default `blockstat_rd_operations`.

The Prometheus metric name for per-disk read operations/second (raw counter, before `rate()`).

`metrics.write_ops`

String, default `blockstat_wr_operations`.

As `metrics.read_ops`, for writes.

`metrics.read_bytes`

String, default `blockstat_rd_bytes`.

The Prometheus metric name for per-disk bytes read (raw counter).

`metrics.write_bytes`

String, default `blockstat_wr_bytes`.

As `metrics.read_bytes`, for writes.

`metrics.read_time_ns`

String, default `blockstat_rd_total_time_ns`.

The Prometheus metric name for cumulative read I/O time in nanoseconds. This is the **primary** load signal by default (`load_weights.iotime`) — see IMPLEMENTATION_PLAN.md section 4 for why I/O time, rather than IOPS or bytes, is the right default.

`metrics.write_time_ns`

String, default `blockstat_wr_total_time_ns`.

As `metrics.read_time_ns`, for writes.

`metrics.labels.vmid`

String, default `vmid`.

The Prometheus label carrying the VM id.

`metrics.labels.device`

String, default `instance`.

The Prometheus label carrying the drive id (`scsi0`, `virtio0`, ...). **Confirm this with `verify-metrics`.** PVE's own tag of this name collides with Prometheus's own `scrape-target` instance label, and many Telegraf configurations rename or overwrite it; `verify-metrics` warns loudly when this is still set to the literal string `instance`.

`metrics.labels.node`

String, default `nodename`.

The Prometheus label carrying the PVE node name.

metrics.rate_window

Duration, default 5m.

The inner range passed to `rate()`. Must be at least four times `metrics.pvestatd_push_interval`, or `rate()` sees too few points to be meaningful — `verify-metrics` measures the *observed* sample spacing of a live series and errors if this rule is violated against reality, not just against the declared `pvestatd_push_interval`.

metrics.step

Duration, default 5m.

The sampling step used for range queries (forecasting, coverage checks). Independent of `metrics.rate_window`, though the two are usually set equal.

metrics.pvestatd_push_interval

Duration, default 60s.

How often PVE's `pvestatd` pushes metrics — a PVE-side setting this tool cannot read from the API, so it must be declared here. `verify-metrics` cross-checks it against the *observed* sample spacing of a live series and errors if they disagree by more than 20%, which is what stops a stale declaration from silently invalidating the `rate_window` rule above.

window — the decision window

This is the period whose load is balanced. It is **not** the amount of history a forecaster needs to fit — see `forecast.model` below and `IMPLEMENTATION_PLAN.md` section 10.1.

window.lookback

Duration, default 24h.

The trailing window each disk's load is reduced over. Must be at least as long as the configured `forecast.model` needs (`forecast.model: holt_winters` at its default `seasonal_periods` needs 48h, which a 24h lookback can never supply — this is rejected at startup, not silently degraded).

window.quantile

Fraction in (0, 1), default 0.95.

The point-estimate quantile: each disk's load is the p95 of its raw signal over `window.lookback`, not the mean, so one traffic spike neither triggers nor suppresses a migration.

window.upper_quantile

Fraction in (0, 1), default 0.99.

The quantile the **optimizer** and the saturation guard actually consume — must be $\geq$ `window.quantile`. Being wrong in the direction of “busier than it looks” costs a slightly suboptimal balance; the other direction risks migrating a disk onto a storage that is about to saturate, which is why the optimizer sees the more conservative bound rather than the point estimate.

window.min_coverage

Fraction in (0, 1], default 0.80.

A disk whose sample coverage over `window.lookback` falls below this fraction has its data rejected for this run; its last known load from `state.json` is used instead, flagged in the report. Never treated as zero — that would silently invite migrations *onto* a busy but under-sampled storage.

load_weights — combining read/write and time/ops/bytes

See IMPLEMENTATION_PLAN.md section 4 for the full blend-and-rescale formula; the summary is that iotime/ops/bytes are blended in normalized space and the whole result is rescaled back onto the in-flight-I/O scale, so $l_d = raw_t(d)$ exactly under the defaults below.

load_weights.iotime

Weight, default 1.0.

Weight on the I/O-time term ($rate(rd_total_time_ns + wr_total_time_ns) / 1e9$), the primary signal by default: it is Little's-law in-flight I/O, additive across disks and self-weighting between large sequential and small random access with no manual read/write tuning needed.

load_weights.ops

Weight, default 0.0.

Weight on the operations/second term. Non-zero only if you want IOPS to influence placement independently of I/O time.

load_weights.bytes

Weight, default 0.0.

Weight on the bytes/second term, for the same reason as load_weights.ops.

load_weights.read_factor

Weight, default 1.0.

Read/write asymmetry applied to the ops/bytes terms only (see load_weights.write_factor). Applying it to the iotime term as well double-counts, since I/O time already reflects the real read/write cost difference — do that only deliberately.

load_weights.write_factor

Weight, default 1.0.

As load_weights.read_factor, for the write side. A write-heavy RAID-6 array, for example, might set this above 1.0 on the ops/bytes terms to express the real cost asymmetry those terms cannot otherwise see.

groups — storage groups**groups[].name**

String, required.

A group's name. A disk may only ever move between storages in its own group; there is no cross-group balancing (IMPLEMENTATION_PLAN.md section 5). Must be unique.

groups[].storages[].id

String, required.

A PVE storage id. A storage may belong to **at most one** group — belonging to two would make a disk's eligible destinations ambiguous, and config validation rejects it.

groups[].storages[].capability_weight

Weight, > 0 , default 1.0.

The storage's **relative** share of the group's load — $u_s = L_s / c_s$ is what actually gets equalized, so a storage with half the spindles of its peers can be given 0.5 and will correctly be assigned half the load. Only the *ratios* within a group matter; the absolute value is meaningless and it is not a capacity.

groups[].storages[].reserve_factor

Weight or `null`, default `null` (inherits `snapshot_reserve.factor`).

Per-storage override of the group-wide snapshot reserve factor, for a storage whose snapshot behaviour genuinely differs from its peers.

groups[].storages[].saturation_load

Positive number or `null`, default `null`.

The storage's approximate queue depth — the number of concurrent I/O requests it services before latency climbs super-linearly — in the same units as the load model (average in-flight I/O). Used only by the section 7.3 saturation guard on a migration's mirror. **Has no safe default:** an idle storage's observed load is not its capacity, so leaving this `null` (the default) simply disables that one advisory check for the storage; the hard bounds (`migration.max_single_move_duration`, the transient reserve invariant) always apply regardless. Obtain a real value from the array's documented queue depth, or by observing where latency actually starts climbing.

snapshot_reserve — the free-space floor**snapshot_reserve.factor**

Weight ≥ 0 , default `2.0`.

Keep this many times the largest disk on a storage free at all times, including *during* a migration, not merely before and after — PVE 9's volume-chain snapshots allocate a new full-size volume per snapshot, which is what this protects against. This is the constraint the tool never trades against balance (IMPLEMENTATION_PLAN.md section 5.3, (C5)).

snapshot_reserve.min_free_bytes

Size, default `0`.

An absolute floor, applied as `reserve = max(factor * largest_disk, min_free_bytes)`. Matters when a storage's largest disk is small: with `factor: 2.0` and a 10 GiB largest disk, the snapshot term alone would reserve only 20 GiB on a 20 TiB LUN.

snapshot_reserve.count_foreign_volumes

Boolean, default `true`.

Count volumes DRS does not manage (templates, ISOs, backups, other groups' disks, orphans) against a storage's used capacity. Strongly recommended: turning this off understates real usage and silently erodes the reserve it is meant to guarantee.

gates — deciding whether to act at all**gates.drift_threshold**

Fraction, default `0.10`.

Re-planning is skipped until the load vector has moved by this much in L1, relative to the vector recorded at the last executed balance. Lower reacts sooner and migrates more; below roughly `0.05` the tool will start chasing noise on a busy cluster. Interacts with `gates.imbalance_threshold`: drift decides *whether to look*, imbalance decides *whether to act*.

gates.imbalance_threshold

Fraction, default `0.20`.

The minimum relative spread across a group's storages before a plan is actually built. A group under this threshold is left alone even if it has drifted.

gates.cooldown_per_disk

Duration, default 24h.

A disk that moved within this long is pinned in place for planning purposes. Prevents a disk from being shuffled back and forth across two still-noisy storages.

gates.cooldown_per_storage

Duration, default 1h.

A storage involved in a migration within this long accepts no new incoming moves. **Must exceed the implied wipe time of that storage's largest disk** (`largest_disk / saferemove_throughput`) or the next run will plan onto a storage that is still draining and stall — `pve-storage-drs verify-storages` computes this and warns when the configured cooldown is too short.

migration — cost, bandwidth and the payback rule**migration.bwlimit_bytes_per_sec**

Size/s, default 209715200 (200 MiB/s).

Passed to `move_disk` as `bwlimit` (converted to KiB/s at the API call site — the API's own unit, never bytes/s). Also the divisor in the section 7.1 mirror-duration estimate used by the payback test.

migration.source_load_weight

Weight, default 1.0.

Additional in-flight I/O charged to the **source** storage while a mirror is running — a mirror is one sequential reader, so 1.0 is the natural value in the same units as the load model.

migration.target_load_weight

Weight, default 1.0.

As `migration.source_load_weight`, for the target (one sequential writer).

migration.payback_horizon

Duration, default 7d.

The horizon H over which a plan's imbalance reduction is assumed to persist — $\text{benefit} = \Delta E * H$. Setting this to 0 disables the payback test entirely (`IMPLEMENTATION_PLAN.md` section 11.1 rejects that at config-load time: `payback_horizon > 0` is required).

migration.payback_ratio

Weight > 0 , default 10.0.

λ : a plan is only accepted if its total benefit is at least this many times its total migration cost. This is the numeric form of “migrating a very large disk might generate more traffic than it saves.”

migration.max_single_move_duration

Duration, default 6h.

A hard per-move ceiling on `duration_mirror + duration_wipe`. A move predicted to take longer than this is rejected outright regardless of the aggregate payback test.

migration.account_saferemove_wipe

Boolean, default true.

Include the post-move `saferemove` zeroing pass in cost and duration estimates. PVE's LVM `saferemove` defaults to 10 MiB/s, so a 1.5 TiB disk's cleanup (~44h) can dwarf its ~2.2h mirror; disabling this only if

you have verified `saferemove` is off on every storage in a group understates real migration cost badly if that assumption is wrong.

migration.wipe_load_weight

Weight, default 1.0.

In-flight I/O charged to the source for the whole `saferemove` wipe duration — both in the cost model and in the section 7.3 saturation guard, where a draining move charges this to its source and nothing to its target. The zeroing pass is one sequential writer, so 1.0 is the natural value.

migration.saturation_ceiling

Fraction in (0, 1], default 0.85.

The fraction of a storage’s `saturation_load` (not of `capability_weight`) a move may drive it to. Inactive for any storage whose `saturation_load` is null — the default, since there is no safe way to infer it.

migration.assume_thick_provisioning

Boolean, default true.

Cost and reserve arithmetic use each disk’s *provisioned* size rather than its currently allocated size. Set false only for genuinely thin-provisioned storage, and note that allocation can *grow* during a move even then.

objective — the solver’s trade-off weights

See `IMPLEMENTATION_PLAN.md` section 5.4 for the full objective and section 14.3 for a worked demonstration of `beta_move_count` and `kappa_vm_affinity` choosing between competing plans.

objective.spread_metric

l1 or minmax, default l1.

How imbalance is measured: l1 (sum of each storage’s deviation from the group’s target utilization) or minmax (only the single hottest storage). l1 is the default because minmax is indifferent to a second nearly-as-bad storage once the worst one is fixed.

objective.alpha_spread

Weight, default 1.0.

Weight on the imbalance term. This and the other three weights below share one scale — see the worked arithmetic in `IMPLEMENTATION_PLAN.md` section 14.3 for what “a 0.25-request reduction” actually costs against one move.

objective.beta_move_count

Weight, default 0.25.

Penalty per migration — the tunable implementation of “minimize the number of migrations” (`IMPLEMENTATION_PLAN.md`’s “Requirement interpretation” note: a strict minimum would refuse a second cheap move that halves the remaining imbalance, which is not desired). Raising it trades balance quality for fewer, larger moves.

objective.gamma_move_bytes_per_tib

Weight per TiB, default 0.05.

Penalty per TiB actually migrated, biasing against moving large disks specifically (as opposed to `beta_move_count`, which only counts moves).

objective.kappa_vm_affinity

Weight, default 0.50.

Penalty per extra storage a VM's disks are spread across, counted only **within** a group (a VM split across two groups is structural and cannot be repaired by any migration, so it is not counted). A soft preference: a strong imbalance or a capacity constraint can legitimately override it.

objective.affinity_counts_pinned_disks

Boolean, default false.

Whether a disk that cannot move this run (snapshot-blocked, excluded, or on a locked VM) still counts toward kappa_vm_affinity. Default false so one unreachable disk cannot veto good placement of the rest of its VM's disks; note efidisk0/tpmstate0 are *not* in this pinned set on PVE 9.2 — they move online.

objective.reserve_violation_penalty

Weight, default 1000.0.

A **floor**, not the value actually used, for the single-stage big-M fallback solve path (IMPLEMENTATION_PLAN.md section 5.3, option 2): the engine computes a provably-dominant P from the group's own load and disk sizes at solve time and uses `max(configured, computed)`, warning when it had to raise it. The default lexicographic two-stage solve (option 1, and the default) needs no penalty at all and is unaffected by this key.

solver — which backend plans**solver.backend**

One of auto, cpsat, cbc, heuristic; default auto.

auto prefers CP-SAT, falls back to CBC (the packaged solver path via python3-pulp + coinor-cbc), then the dependency-free heuristic. Force a specific backend only to reproduce or compare a result — heuristic is also the automatic choice for a group too large for either MILP backend to solve within solver.time_limit_seconds.

solver.time_limit_seconds

Duration, default 60.

Wall-clock budget for a MILP solve attempt before falling back to the heuristic. Never emits a partial/unvalidated assignment on timeout — it falls back cleanly instead.

solver.mip_gap

Fraction, default 0.02.

Acceptable optimality gap for the MILP solve. IMPLEMENTATION_PLAN.md section 5.5's coefficient-scaling error bound is three orders of magnitude below this, so it cannot itself change which plan is selected.

solver.heuristic_iterations

Integer > 0, default 5000.

Iteration budget for the heuristic's local-search descent phase (section 5.5). Higher can find a better local optimum on a large group at the cost of run time.

execution — how (and whether) moves actually happen**execution.mode**

One of dry-run, confirm, auto; default dry-run.

dry-run (the default) prints the plan and changes nothing. confirm executes one move at a time with a prompt before each. auto executes unattended, subject to the concurrency and time-window settings

below. `--mode` on the command line overrides this for one run; moving *toward* less safety is logged at warning level (`IMPLEMENTATION_PLAN.md` section 11.3).

execution.max_concurrent_migrations

Integer $\geq$ 1, default 1.

How many moves may be in flight across the whole run at once. Above 1 requires the *generalized* transient reserve invariant (section 8.1): several disks can land on one storage at once, and none of their sources release space until each individually completes.

execution.max_migrations_per_run

Integer $\geq$ 1, default 5.

A ceiling on how many moves one invocation executes, independent of how many the plan contains — the remainder waits for the next run.

execution.max_concurrent_per_storage

Integer $\geq$ 1, default 1.

Caps concurrent moves touching one storage, counting it as either source or target. At the default of 1, two moves targeting the same storage serialize automatically and the *generalized* transient invariant collapses to its simple single-move form. Raising it should be a deliberate act on a storage with real spare headroom — two moves off the same source also means two concurrent `saferemove` wipes sharing one throttle.

execution.max_replans_per_run

Integer $\geq$ 0, default 3.

A cap on how many times one run may abandon its current plan and re-plan from newly observed state (section 9.2) — a mismatch between the plan and reality (a VM live-migrated mid-plan, a lock appeared) is normal, but a cluster churning faster than the engine can plan is a condition for a human, not for indefinite retrying.

execution.abort_on_failure

Boolean, default `true`.

On a failed move, stop the run and report rather than continuing with the remaining plan. The cluster is left in a valid intermediate state either way; the next run re-plans from reality.

execution.poll_interval_seconds

Duration, default 10.

How often an in-flight `move_disk` task's status is polled.

execution.locks.wait_timeout

Duration, default 4h.

How long to wait for a VM's config lock (any non-empty value — the set is treated as open-ended and never whitelisted) to clear before applying `execution.locks.on_timeout`. Generous by default: a backup of a large VM can easily run for hours, and that is a normal, not exceptional, condition.

execution.locks.poll_interval

Duration, default 30s.

How often the lock is re-checked while waiting.

execution.locks.on_timeout

One of skip, abort; default skip.

What happens when `execution.locks.wait_timeout` elapses: skip this move and continue with the rest of the plan, or abort the run. Never “force” — that option does not exist, deliberately.

execution.source_release.wait

Boolean, default true.

Wait for the source volume of a completed move to actually disappear (and the VM to be unlocked) before considering the move done, rather than trusting the `move_disk` task’s own OK status — with `saferemove` enabled, OK does not mean the source space is back yet. Set false only on storages verified not to wipe.

execution.source_release.timeout

Duration, default 48h.

Bound on the wait above, sized for a multi-TiB disk wiping at the default 10 MiB/s. Must be at least `largest_disk / saferemove_throughput` for every storage where `saferemove` is on — `verify-storages` checks this.

execution.time_windows[].days

List of mon..sun, default: every day.

Which days a time window applies on. Applies only to `execution.mode: auto` — `dry-run` and `confirm` are not time-restricted, since neither changes anything without a human already present.

execution.time_windows[].start / execution.time_windows[].end

HH:MM strings, required together, must differ.

The window auto mode is permitted to execute moves in. A window may cross midnight (start: "22:00", end: "06:00"); `start == end` is rejected as ambiguous — it would silently mean either “never” or “always”.

exclude — what DRS never touches**exclude.vmid**

List of integers, default [].

VM ids whose disks are never moved.

exclude.disks

List of "vmid:device" strings, default [], e.g. ["101:scsi1"].

Individual disks excluded regardless of their VM.

exclude.storages

List of storage ids, default [].

Storages excluded as both source and target — effectively removes them from their group without editing groups.

exclude.tags

List of PVE tags, default ["no-drs"].

A VM carrying any of these tags opts out entirely.

exclude.skip_vms_with_snapshots

Boolean, default true.

Pre-filter disks with a snapshot or an unreferenced companion volume before they reach the API. Setting this false does **not** make such a move work — PVE rejects `move_disk delete=1` on a snapshotted volume regardless; it only removes the early, informative warning. Keep this true.

exclude.running_only

Boolean, default true.

Only running VMs generate load to balance; a stopped VM's disks are moved only if a capacity constraint requires it. Setting this false widens the movable set to stopped VMs' disks for capacity purposes but they still contribute zero load.

exclude.include_unused_disks

Boolean, default true.

Whether detached (unusedN) volumes are movable. They carry zero load, so the solver only relocates one to repair a capacity violation — never for balance, since there is no benefit to buy. Set false to pin them in place entirely; they still count via `snapshot_reserve.count_foreign_volumes` either way.

report**report.warn_pinned_load_fraction**

Fraction in (0, 1], default 0.25.

If pinned load (snapshots, locks, exclusions) exceeds this fraction of a group's total, the residual imbalance may be structural rather than a planning failure, and the report says so alongside the best achievable spread given the pins.

state**state.path**

File path, default `/var/lib/pve-storage-drs/state.json`.

Local disk, one copy per host, deliberately never `/etc/pve` — see [00-installation.md](#). Losing this file is safe but not free: cooldowns and the drift baseline reset, so the next run may migrate sooner than intended. Treat it as state to back up, not as a cache.

`show-load` and `plan` both read this file (if present) for the drift gate's history; neither writes it, and a missing or unreadable file just means every group evaluates as if it had never been balanced before — see [../internals/15-state.md](#).

forecast — history beyond the plain quantile

See `IMPLEMENTATION_PLAN.md` section 10 and `proxmox_storage_drs/forecast.py`.

forecast.model

One of `quantile`, `seasonal_naive`, `holt_winters`; default `quantile`.

`quantile` needs only `window.lookback` and no model fitting. `seasonal_naive` and `holt_winters` need more history than the decision window alone — see below — and `pve-storage-drs` refuses to start if `window.lookback` (or your Prometheus retention) cannot supply it, rather than silently falling back.

forecast.seasonal_lookback_days

Days, default 7.

History seasonal_naive needs: same-hour-of-day samples across this many days. 0 disables seasonal_naive (falls back to quantile, logged).

forecast.holt_winters.seasonal_periods

Integer > 0, default 288 (24h at a 5-minute step).

Samples per seasonal cycle. Combined with metrics.step, this determines holt_winters's required history: $2 * \text{seasonal_periods} * \text{metrics.step} - 48\text{h}$ at the defaults, which a 24h window.lookback can never satisfy.

forecast.holt_winters.trend

One of add, mul, none; default add.

The trend component passed to the underlying Holt-Winters fit (statsmodels, an optional dependency — see [30-safety-and-status.md](#)).

forecast.holt_winters.seasonal

One of add, mul, none; default add.

The seasonal component passed to the same fit.

forecast.holt_winters.residual_z

Weight, default 2.0.

The upper bound is $\text{point_estimate} + \text{residual_z} * \text{stdev}(\text{residuals})$ from the in-sample fit — this is what the optimizer actually consumes (see window.upper_quantile for the equivalent under the quantile model).

Verifying your setup

Run this before relying on any plan. Telegraf and InfluxDB naming varies by deployment, so pve-storage-drs never assumes the defaults in [10-configuration.md](#) are correct for your cluster — verify-metrics is what checks them against the live Prometheus instead.

Running it

```
pve-storage-drs -c /etc/pve/drs.yaml verify-metrics
```

A successful run looks like this (labels and values are illustrative):

```
[ info] metrics.read_ops = 'blockstat_rd_operations' exists
[ info] metrics.write_ops = 'blockstat_wr_operations' exists
[ info] metrics.read_bytes = 'blockstat_rd_bytes' exists
[ info] metrics.write_bytes = 'blockstat_wr_bytes' exists
[ info] metrics.read_time_ns = 'blockstat_rd_total_time_ns' exists
[ info] metrics.write_time_ns = 'blockstat_wr_total_time_ns' exists
[ info] blockstat_rd_operations: sample series labels {'vmid': '101', 'instance': 'scsi0', 'nodename': 'pve01'}
...
OK: verify-metrics found no blocking problems
```

Exit status is 0 when every check passes, 1 when any check reports an error-level finding (see below). --json emits the same information as one JSON object instead — see [30-safety-and-status.md](#) for exit codes in general.

What each finding means

verify-metrics runs six checks, in this order (IMPLEMENTATION_PLAN.md section 3.3):

1. **Metric names exist.** Each of `metrics.read_ops`, `metrics.write_ops`, `metrics.read_bytes`, `metrics.write_bytes`, `metrics.read_time_ns` and `metrics.write_time_ns` is looked up against Prometheus's own `/api/v1/label/___name___/values`. A missing name is an **error** — fix the name in `10-configuration.md`'s `metrics.*` section to match what your Telegraf actually emits.
2. **Sample series and labels.** One instant query per metric, printing its full label set so you can see at a glance whether `metrics.labels.vmid`, `metrics.labels.device` and `metrics.labels.node` are really present and non-empty on a live series. A metric that exists but currently has no series is a **warning**, not an error — it may simply mean nothing has generated that kind of I/O recently.
3. **Configured labels present.** If `metrics.labels.vmid`/`metrics.labels.device`/`metrics.labels.node` do not appear (or are empty) on the sample series, that is an **error**: the load model has nothing to join disks on.
4. **The instance collision.** If `metrics.labels.device` is still the literal string `instance`, a **warning** — PVE's own per-disk tag has this name and it collides with Prometheus's unrelated scrape-target instance label. Many Telegraf configurations rename or overwrite one of the two; confirm which one survived before trusting this.
5. **Per-disk coverage.** For every (`vmid`, `device`) pair seen on `metrics.read_ops`, the fraction of expected samples actually present over `window.lookback` is measured. Below `window.min_coverage`, a **warning** naming the disk — that disk's own data will be rejected by the load model at plan time and its last known load from `state.json` used instead.
6. **Observed sample spacing.** The modal gap between consecutive samples of a live series is measured directly, rather than trusted from `metrics.pvestatd_push_interval`. Disagreeing by more than 20%, or `metrics.rate_window` being below four times what was actually observed, is an **error** — this is what stops a stale `pvestatd_push_interval` declaration from silently breaking the `rate()` window it is supposed to protect.

If it fails

- **A metric does not exist:** check your Telegraf `influxdb_listener` (or equivalent) configuration for the actual measurement/field names it is producing, and update `metrics.*` to match.
- **A label is missing or empty:** the sample series printed by check 2 is the fastest way to see what labels Prometheus actually has for that series — update `metrics.labels.*` to name the real ones.
- **Coverage is low for specific disks:** check that Telegraf is actually scraping every node, and that `window.lookback` is not longer than your Prometheus retention.
- **Spacing disagrees:** measure `pvestatd`'s real push interval on a node (its `systemd` timer, or the interval visible in `/etc/pve/status.cfg` if set there) and correct `metrics.pvestatd_push_interval`.

Reading show-load and verify-storages

Both commands only read — neither ever changes anything, in any `execution.mode`. Run them any time you want to see what the tool sees.

show-load

Prints every storage in every configured group, its disks, and its snapshot-reserve status. This example uses the group from IMPLEMENTATION_PLAN.md section 14's worked example (`config/drs.example.yaml` ships the same group and weights) so the numbers are traceable to that section rather than invented for this page:

```
$ pve-storage-drs -c /etc/pve/drs.yaml show-load
```

```
Group fc-tier1 → ACT: reserve violated on san-a; bypassing the drift and imbalance gates (section 13: safety is not subject to hysteresis)
```

```

san-a  used 4.50 TiB/8.00 TiB  L=6.50 u=6.50  ▲ reserve short by 512.00 GiB  (largest disk 2.00 TiB,
requires 4.00 TiB free)
  101:scsi0      2.00 TiB  raw    ℓ 3.00
  101:scsi1      1.00 TiB  raw    ℓ 1.00
  102:scsi0      1.50 TiB  raw    ℓ 2.50
san-b  used 1.50 TiB/8.00 TiB  L=0.70 u=0.70  reserve OK  (largest disk 1.00 TiB, requires 2.00 TiB free)
  103:scsi0     512.00 GiB  raw    ℓ 0.40
  104:scsi0      1.00 TiB  raw    ℓ 0.30
san-c  used 512.00 GiB/8.00 TiB  L=0.20 u=0.20  reserve OK  (largest disk 512.00 GiB, requires 1.00 TiB
free)
  105:scsi0     512.00 GiB  raw    ℓ 0.20

```

L=/u= on a storage's line are section 4's L_s (summed ℓ of the disks currently on it) and u_s = L_s / capability_weight; ℓ after a disk's size/format is that disk's own share. Both come from IMPLEMENTATION_PLAN.md section 14's worked example, so the numbers above are traceable to that section rather than invented for this page.

The Group <name> → ACT/NO ACTION line

Section 6's three gates, evaluated in order — reserve override, then drift, then imbalance — and the first one that decides wins. The header line above shows the **reserve override**: san-a's 0.5 TiB shortfall forces ACT outright, regardless of how balanced or drifted the group is, because "safety is not subject to hysteresis" (section 13). The other two shapes this line can take, on a group with no reserve violation:

```

Group fc-tier1 → ACT: imbalance 255.4% meets or exceeds gates.imbalance_threshold (20.0%)
Group fc-tier1 → NO ACTION: imbalance 12.2% is below gates.imbalance_threshold (20.0%)

```

show-load reads state.path's recorded load vector (section 11.2) and passes it to the drift gate, so once a group has one on record, this line can also read:

```

Group fc-tier1 → NO ACTION: drift 3.1% is below gates.drift_threshold (10.0%)

```

A group with no state.json, or none recorded for it yet, still evaluates as if this were the very first run — the drift gate is skipped outright (section 6's own degenerate-case rule), not "treated as zero drift" — so its verdict falls straight through to a reserve override or an imbalance check, exactly as before this was wired up. docs/internals/80-gates.md and docs/internals/15-state.md have the detail; nothing yet writes last_balance (that needs execute.py, not yet written), so this stays the common case until a migration has actually run.

A pinned disk carries [pinned: <reason>] after its size and format — snapshots present (N), locked: <lock>, excluded by config, or excluded: unused disk (exclude.include_unused_disks=false), matching IMPLEMENTATION_PLAN.md section 5.3 (C2) exactly; that reason is the answer to "why won't it move this disk."

A disk whose measured I/O falls below window.min_coverage (section 3.4) gets a ▲ <disk> : <reason> line of its own, right after that group's disks, instead of a silently wrong ℓ — it either shows a state.json-recorded last known load (see docs/internals/70-loadmodel.md) or 0.0 when no state.json entry exists for it, and either way the warning says which. tpmstate0 and unusedN disks are never flagged this way: they genuinely emit no I/O metrics, so ℓ 0.00 for them is correct, not a gap. A group with no measured I/O at all gets one (idle: no measured I/O for this group this window) line instead of per-disk warnings.

A Prometheus outage does not fail this command. Sizes and reserve status never depend on Prometheus at all; if the load fetch for a group fails, that group's storage/disk lines simply omit L=/u=/ℓ, and a ▲ per-disk load unavailable: <reason> line explains why — check verify-metrics first if you see this.

A **Warnings:** block, when present, lists things worth a look but not fatal: a disk on a storage that is not in any configured group (“ungrouped, not managed” — check whether that storage belongs in a group), or a disk whose size came from its own config rather than the storage’s authoritative content listing. The latter usually means the volume no longer exists on the storage — most often because it was removed directly on the storage backend rather than through Proxmox, which leaves a dangling reference behind that PVE does not clean up or even notice on its own (a VM starts fine with one; see IMPLEMENTATION_PLAN.md section 3.6 for a confirmed case). If you see this warning, check whether the named volume still exists on the storage and, if it genuinely does not, remove the stale reference (`qm unlink <vmid> <device>` for an unusedN entry).

--json emits the same information as one object with `groups[].storages[]` (including the exact byte counts behind the reserve check, plus load and utilization when available) and `groups[].disks[]` (plus load and `load_flagged_reason` when available), with each group carrying its own `load_computed`, `idle`, `load_error` and `gate` fields — a Prometheus outage on one group never prevents another group’s `load_computed: true` or `gate` from being reported. `gate` is `null` when `load_computed` is `false` (no `GroupLoad` to evaluate gates against) and otherwise an object with `act` (bool), `reason` (string, identical to the human line’s text after the arrow), `reserve_override` (bool), and `drift_fraction/imbalance_fraction` (float or `null` — `null` means that gate was never reached, not that it evaluated to zero).

A config with several groups issues Prometheus queries per group. Computing one group’s load takes seven queries (six raw metrics plus one coverage check), unfiltered by group — the fastest correct way to get one group’s numbers, but it means a config with N groups makes 7N queries in total on every `show-load`, not 7, since each group’s coverage and data-quality decisions are independent and so cannot share a single fetch. For the typical one-to-three-group deployment this is a handful of fast instant queries and not worth worrying about; an operator running many groups against an already-busy Prometheus should be aware of the multiplier.

verify-storages

Reports `saferemove` and the wipe time it implies for the largest disk on each storage, and warns when your configured cooldown or move-duration limits are shorter than that implied wipe — the condition IMPLEMENTATION_PLAN.md section 9.3 describes as “the next run plans onto a storage that is still draining”. Section 14’s fixture has `saferemove` off everywhere (stated explicitly there, so its payback arithmetic is reproducible); this example instead assumes an LVM `san-b` with `saferemove` on at PVE’s default 10 MiB/s throughput, to show what the warning looks like:

```
$ pve-storage-drs -c /etc/pve/drs.yaml verify-storages
Group fc-tier1
  san-a  saferemove=off
    saferemove is off or throughput unknown; no wipe-time check
  san-b  saferemove=on
    implied wipe time for the largest disk (1.00 TiB): 1.2d
    ▲ gates.cooldown_per_storage (1.0h) is shorter than the implied wipe time -- the next run may plan onto
    a still-draining storage (section 9.3)
    ▲ migration.max_single_move_duration (6.0h) is shorter than the implied wipe time -- a move of the
    largest disk would be rejected outright (section 7.3)
  san-c  saferemove=off
    saferemove is off or throughput unknown; no wipe-time check
```

Storage types without `saferemove` at all (Ceph RBD, ZFS) always report `saferemove=off` and skip the check — there is nothing to wipe.

Reading plan

plan computes what pve-storage-drs would do and prints it. It never changes anything, in any execution.mode — that is apply’s job (not yet implemented; see 30-safety-and-status.md), and it needs a separate, explicit invocation to run at all.

For each group, plan:

1. Fetches the same load ($\ell_d/L_s/u_s$) show-load does, and evaluates the same section 6 gate.
2. If the gate says NO ACTION, stops there for that group — no solver runs, nothing more to show.
3. If it says ACT, runs the heuristic solver (IMPLEMENTATION_PLAN.md section 5.4/5.5) to compute a target assignment, orders its moves under the section 8 transient reserve invariant, and checks the whole plan against section 7’s payback rule.

This example uses the same section 14 worked example show-load’s manual page does, at the default weights (the three-move plan), so the numbers are traceable to that section:

```
$ pve-storage-drs -c /etc/pve/drs.yaml plan
Group fc-tier1 → ACT: reserve violated on san-a; bypassing the drift and imbalance gates (section 13: safety
is not subject to hysteresis)
  1. 102:scsi0      san-a → san-c      1.50 TiB   ~2.2h   Δimbalance -4.53   ℓ/z 1.67
  2. 101:scsi1      san-a → san-b      1.00 TiB   ~1.5h   Δimbalance -2.00   ℓ/z 1.00
  3. 105:scsi0      san-c → san-b     512.00 GiB ~43.7m   Δimbalance -0.40   ℓ/z 0.40
after: san-a=3.00  san-b=1.90  san-c=2.50
spread: 255.4% → 44.6%
payback: benefit 4.19e+06 load·s vs cost 3.15e+04 load·s → ratio 133 (need 10) ✓
```

The after:/spread: lines (and the payback numbers) reflect what the scheduler actually managed to order, not the solver’s target assignment — identical here since all three moves scheduled cleanly, but see the deadlock note below for when a plan cannot schedule everything it proposed.

A group the gate does not act on prints one line and stops:

```
Group fc-tier2 → NO ACTION: imbalance 0.0% is below gates.imbalance_threshold (20.0%)
```

Reading a move line

1. 102:scsi0 san-a → san-c 1.50 TiB ~2.2h Δimbalance -4.53 ℓ/z 1.67 — the disk, its current and target storage, its size, the estimated mirror duration (size / migration.bwlimit_bytes_per_sec, plus a +wipe <time> suffix when saferemove on the source storage adds one — see the payback section below), this move’s own effect on the group’s imbalance metric at the moment it was scheduled (negative means imbalance went down, which is the usual case; a move scheduled mainly to resolve a reserve violation or consolidate a VM can show a positive value and still be correct), and ℓ/z — load per TiB, section 7.3’s own “single best indicator of a good migration candidate: high I/O concentrated in a small disk.”

A move whose own duration exceeds migration.max_single_move_duration carries a Δ exceeds migration.max_single_move_duration suffix and always fails the plan (see below) — this is a hard, per-move rule, independent of whether the plan as a whole looks profitable.

resolves_reserve_violation (visible in --json, and implied by the ACT: reserve violated on ... header in human output) marks a move scheduled first *regardless* of its ratio, because the storage it leaves is currently breaching (C4)/(C5) — safety is not subject to hysteresis (section 13), so this always wins over a purely balance-driven move.

A Δ line means a deadlock, not a hidden failure. If the target assignment includes a move this run cannot find any transient-feasible order for, plan says so explicitly (IMPLEMENTATION_PLAN.md section 8.3’s “report, never force” path — staging and plan-splitting are not yet implemented, so this is where

a genuine cycle or an unavoidable capacity shortfall currently surfaces) rather than silently omitting the move or, worse, telling you a plan is clean when part of it is not achievable. When only *some* moves deadlock, the `after:/spread:/payback:` lines report the state reachable by the moves that did schedule, never the fuller picture the undeliverable ones would have produced.

The payback: line

Section 7.3's acceptance test: $\text{benefit} \geq \text{migration.payback_ratio} * \text{cost}$, both sides in load-seconds. benefit is $(E_{\text{before}} - E_{\text{after}}) * \text{migration.payback_horizon}$ (section 7.2's own *unweighted* E , not the solver's α_{spread} -scaled objective term — the two agree numerically at the default $\alpha_{\text{spread}}: 1.0$, but the payback ratio must not depend on that tuning knob); E_{after} is evaluated against what plan's own scheduler actually managed to order, not the solver's aspirational target, so a partially-deadlocked plan is scored on the moves it can really make, not ones it cannot. cost sums every *scheduled* move's $\text{duration_mirror} * (\text{source_load_weight} + \text{target_load_weight}) + \text{duration_wipe} * \text{wipe_load_weight}$ (section 7.1). A ✓ plan passed; a ✗ one did not.

A ✗ plan can fail for two different reasons, reported as two different lines, because they call for different fixes:

- **Economic failure** — the balance benefit does not outweigh the migration cost. Adjust objective's weights, `migration.payback_ratio`, or accept the plan is not worth running.
- **Hard-duration failure** — some move's own `duration_mirror + duration_wipe` exceeds `migration.max_single_move_duration`, regardless of whether the plan as a whole is profitable. Fix `saferemove.throughput`, `migration.bwlimit_bytes_per_sec`, or the duration limit itself.

Either, both, or neither can apply to the same plan; plan names exactly which happened rather than one generic “does not pass the payback test” line for both.

A plan resolving a reserve violation always passes this test, regardless of the ratio shown — the example above happens to pass on merit ($133 \geq 10$), but a plan whose *only* move fixes a (C4)/(C5) violation with zero balance benefit (a real, common case: relocating the sole loaded disk in a two-storage group changes which side carries it without reducing spread at all) is accepted too. Section 13's “the reserve is never traded against balance” applies here exactly as it does to the gates: an operator does not get to decline a capacity emergency fix because it scores poorly against `migration.payback_ratio`. The hard per-move duration rule is not exempted this way — it is an operational limit, not an economic one, and still blocks the plan.

What a ✗ (or a rejected move) does *not* do today: automatically make the plan smaller and retry.

Section 7.3 describes re-solving with β and γ doubled, up to three times, converging on the highest-value subset of moves. That loop is not implemented — a failing plan is reported and left for you to review, not silently adjusted. See `docs/internals/96-payback.md`.

What plan does not yet do

- **No automatic re-solve-and-shrink on a failing payback test** (see above) — reported, not fixed for you.
- **No section 7.3 saturation-ceiling defer check**. Needs a forecaster upper bound this codebase does not compute yet, and no group in this project's own dogfooding cluster has `saturation_load` set — which the plan itself says is “fully supported... loses only this one advisory check.” `max_single_move_duration` and the transient reserve invariant are the two *hard* bounds and are both already enforced.
- **state.json drift history is read, never written**. plan's gate now reads real `last_balance` history when a group has one recorded (see `docs/internals/15-state.md`), but nothing writes `last_`

balance yet – that needs `execute.py` (phase 7), so a group stays “first run” (drift gate skipped) until a migration has actually run.

- **No cooldowns, no staging, no concurrent scheduling.** Cooldown timestamps are stored and round-tripped by `state.py` but not yet *read* by `topology.py`’s (C2) pinning or `heuristic.py`’s target eligibility. All documented as deliberate, not forgotten, in `docs/internals/15-state.md`, `docs/internals/90-heuristic.md` and `docs/internals/95-schedule.md`.

`--json` emits `groups[]`, each with `gate` (identical shape to `show-load`’s), `moves[]` (`disk_key`, `vmid`, `device`, `from_storage`, `to_storage`, `size_bytes`, `imbalance_reduction`, `resolves_reserve_violation`, `load_per_tib`, `duration_mirror_seconds`, `duration_wipe_seconds`, `cost_load_seconds`, `exceeds_max_duration`), `deadlocked` (a list of disk keys) and `deadlock_message` (null if none), `before_spread/after_spread` (the section 6 spread fraction, before the plan and after every scheduled move), `load_error` (null unless Prometheus failed for this group), and `payback` — null when there is no `GroupLoad` or the gate said `NO ACTION`, otherwise an object with `benefit_load_seconds`, `total_cost_load_seconds`, `ratio`, `aggregate_ok` (the economic test alone, or true if exempted), `rejected_moves` (disk keys failing the hard duration rule) and `accepted` (`aggregate_ok` and no rejected move).

Safety properties, exit codes, and what this build actually does

What is safe, unconditionally

These hold regardless of `execution.mode`, and are not configurable away (`.agents/domain-invariants.md`):

- **Dry-run is the default.** A missing or unparseable `execution.mode` means dry-run, never automatic execution.
- **The snapshot reserve is never traded for balance.** `pve-storage-drs` will leave a group imbalanced rather than breach `snapshot_reserve.factor/min_free_bytes` on any storage, and that reserve holds *during* a migration, not merely before and after it.
- **Nothing is ever deleted automatically.** An orphaned volume left by a failed move is reported, never removed. The operator deletes.
- **An explicitly named configuration that cannot be read or fails validation is a hard failure,** never a silent fallback to `/etc/pve/drs.yaml`.

Exit codes

Code	Meaning
0	Success. A run that stopped at a gate, found nothing worth moving, or (for <code>verify-metrics</code>) found no error-level finding.
1	The run failed: invalid configuration, an unreachable Prometheus or PVE API, a failed check, or a command not yet implemented in this build (see below).
2	Usage error on the command line.

What this build actually implements

`IMPLEMENTATION_PLAN.md` section 12 lays out nine phases. This is a work in progress; being honest about exactly where it stands matters more than a document that reads as if the tool were finished:

Command	Status
<code>--version</code> , <code>--help</code> , <code>--manual</code>	Implemented.
<code>verify-metrics</code>	Implemented: all six <code>IMPLEMENTATION_PLAN.md</code> section 3.3 checks, human and <code>--json</code> output.

Command	Status
show-load	Implemented: every storage in every group, its disks (all buses), sizes, reserve status ((C4)/(C5)), per-disk/per-storage I/O load ($\ell_d/L_s/u_s$, section 4), and a section 6 act/no-act verdict per group with its reasoning, pinned and low-coverage disks flagged with their reason, human and --json output. A Prometheus outage degrades this one group's load (and its gate verdict) to "unavailable" rather than failing the whole command — sizes and reserve status are unaffected. The gate verdict now reads real drift history from state.json when a group has one recorded; a group with none yet still evaluates as a first run (reserve override or imbalance only, never drift-suppressed) — see docs/internals/15-state.md and docs/internals/80-gates.md.
verify-storages	Implemented: saferemove and the implied wipe time per storage, warning when gates.cooldown_per_storage or migration.max_single_move_duration is shorter than it.
plan	Implemented via the heuristic backend (section 5.5): per group, evaluates the section 6 gate (reading real drift history from state.json when a group has one recorded — same as show-load, see docs/internals/15-state.md), and on ACT runs the heuristic solver, orders its moves under the section 8 transient reserve invariant, and checks the whole plan against section 7's payback (cost/benefit) test — a plan resolving a reserve violation is exempt from the economic half of that test but not from the hard per-move duration rule. Human and --json output. No automatic re-solve-and-shrink on a failing payback test — a plan that fails is reported, not silently adjusted (section 7.3's 3-retry loop is not implemented). No section 7.3 saturation-ceiling check — needs a forecaster upper bound this codebase does not compute yet; fully supported to be absent per the plan's own words, since it is the one advisory (not hard) check. The MILP backend (optimize.py, phase 6) does not exist, so solver.backend has no effect yet — see docs/manual/27-plan.md and docs/internals/90-heuristic.md/95-schedule.md/96-payback.md.
explain	Not implemented yet — needs the payback arithmetic it exists to explain (phase 5).
apply	Not implemented yet — needs execute.py (phase 7 onward).

Configuration loading and validation (this whole manual's [10-configuration.md](#)) is complete and exercised by every command, including the ones above that are not yet implemented: a config error is reported and the run exits 1 before reaching the "not implemented" message, so validating a configuration file does not require waiting for the rest of the engine.

Running a command that is not implemented yet prints a message naming the plan section it belongs to, and exits 1:

```
$ pve-storage-drs apply
pve-storage-drs: 'apply' is not implemented yet in this development build; see IMPLEMENTATION_PLAN.md
section 12 for the phase it belongs to
```

Optional dependencies

pve-storage-drs runs, plans and (once implemented) executes with only requests, ruamel.yaml and jsonschema installed. Two dependencies are optional and are imported only where they are used, never at module level:

- **pulp + coinor-cbc** (Debian Recommends) — the packaged MILP solver path. Without it, the solver falls back to the dependency-free heuristic.
- **statsmodels** (Debian Suggests) — needed only for `forecast.model: holt_winters`. Without it, that model logs a warning and falls back to quantile automatically rather than failing.